

RUHNU, Estonia

WMO: 262270

Lat: 57.7833N

Lon: 23.2589E

Elev: 3

StdP: 101.29

Time Zone: 2.00 (EUE)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-13.1	-9.7	-15.8	0.9	-12.5	-12.9	1.2	-8.3	14.0	0.3	12.3	-0.8	4.7	60	0.631

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.7	24.6	19.9	23.2	19.2	21.9	18.4	20.9	23.6	19.9	22.4	18.9	21.3	3.5	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.7	14.5	22.7	18.8	13.6	21.7	17.8	12.8	20.7	60.4	23.7	56.9	22.5	53.7	21.3	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.2	10.7	9.6	-13.1	26.2	4.1	1.3	-16.0	27.2	-18.4	27.9	-20.6	28.7	-23.6	29.6		
(5)			-13.6	21.8	3.7	1.2	-16.3	22.7	-18.4	23.4	-20.5	24.1	-23.2	24.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.7	-1.4	-2.3	0.0	4.4	10.3	14.6	18.2	18.1	14.0	8.7	4.9	2.2		
	DBStd	7.76	4.07	4.39	3.38	2.73	3.44	2.79	2.19	2.17	2.36	3.07	2.92	3.31		
	HDD10.0	1677	354	344	311	171	37	1	0	0	2	61	154	241		
	HDD18.3	3935	613	578	569	419	249	117	29	29	129	298	404	499		
	CDD10.0	838	0	0	0	2	47	138	254	253	123	21	0	0		
	CDD18.3	53	0	0	0	0	1	4	24	23	1	0	0	0		
	CDH23.3	99	0	0	0	0	1	8	59	31	0	0	0	0		
	CDH26.7	4	0	0	0	0	0	0	2	2	0	0	0	0		
(14) Wind	WSAvg	4.6	5.7	5.0	4.1	3.6	3.5	3.8	3.7	4.2	4.5	5.1	6.0	6.4		
(15) Precipitation	PrecAvg	588	44	35	31	31	38	56	64	70	52	65	57	45		
	PrecMax	927	91	73	61	75	88	128	124	226	119	121	98	98		
	PrecMin	271	8	3	7	3	9	12	2	1	5	7	4	19		
	PrecStd	121	23	16	15	19	22	27	35	47	27	28	26	18		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.5	4.7	8.3	15.5	22.2	24.7	26.8	26.3	20.6	15.8	10.4	8.7		
		MCWB	5.7	3.3	4.6	9.1	16.6	18.5	21.1	21.4	17.4	13.7	9.4	7.8		
	2%	DB	5.5	3.6	6.0	12.2	19.6	22.4	25.0	24.2	19.4	14.6	9.6	7.7		
		MCWB	4.7	2.8	3.9	7.5	14.7	18.1	20.1	20.4	16.6	12.9	8.6	6.8		
	5%	DB	4.5	2.8	4.7	10.1	17.5	20.7	23.5	23.0	18.4	13.6	9.0	6.8		
		MCWB	3.8	2.2	3.1	6.6	13.5	17.0	19.3	19.5	15.9	12.2	8.2	6.0		
	10%	DB	3.5	2.2	3.6	8.4	15.6	19.3	22.1	21.8	17.5	12.7	8.4	5.9		
		MCWB	2.8	1.6	2.5	5.7	12.6	16.0	18.6	18.6	15.2	11.5	7.6	5.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.8	3.7	5.2	9.7	17.4	20.0	22.3	22.5	18.0	14.3	9.6	8.1		
		MCDB	6.2	4.2	7.6	13.8	20.9	23.4	25.8	25.0	19.8	15.3	10.1	8.5		
	2%	WB	4.9	3.0	4.1	8.2	15.8	18.7	21.0	21.0	17.1	13.3	8.9	7.0		
		MCDB	5.5	3.4	5.5	11.4	18.6	21.5	23.8	23.6	18.8	14.2	9.3	7.6		
	5%	WB	3.9	2.3	3.3	6.9	14.3	17.3	20.0	20.0	16.2	12.4	8.3	6.1		
		MCDB	4.5	2.7	4.4	9.5	16.7	20.2	22.6	22.4	17.9	13.3	8.8	6.7		
	10%	WB	2.9	1.7	2.6	5.9	12.7	16.2	19.0	19.1	15.5	11.7	7.7	5.3		
		MCDB	3.4	2.1	3.5	8.1	15.1	19.0	21.6	21.3	17.2	12.6	8.3	5.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	2.8	3.0	4.2	5.5	6.5	6.0	5.7	5.2	4.4	3.6	2.6	2.5		
		MCDBR	2.5	2.7	5.0	8.3	8.7	7.8	7.4	6.3	5.4	3.9	2.8	2.3		
	5% WB	MCWBR	2.6	2.4	3.5	4.5	4.8	4.3	3.9	3.5	3.7	3.0	2.7	2.4		
		MCWBR	2.4	2.5	4.6	7.2	7.8	7.1	6.7	5.4	4.8	3.5	2.6	2.2		
(40) Clear-Sky Solar Irradiance	taub		0.296	0.319	0.342	0.378	0.372	0.372	0.402	0.396	0.364	0.345	0.333	0.293		
		taud	2.198	2.200	2.217	2.287	2.410	2.454	2.390	2.405	2.469	2.412	2.247	2.180		
	Ebn at Noon		543	704	797	830	863	867	828	806	780	682	490	426		
		Edh at Noon	51	79	101	112	106	103	107	99	80	65	50	39		
(44) All-Sky Solar Radiation	RadAvg		0.35	0.97	2.46	4.04	5.61	5.90	5.55	4.38	2.81	1.31	0.39	0.23		
	RadStd		0.07	0.18	0.31	0.47	0.49	0.34	0.44	0.39	0.34	0.17	0.07	0.03		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page